

1 The geometry of folding

What makes a crease pattern a valid manifold?

- Developability and why a folded sheet has zero Gaussian curvature on facets.
- Rigid vs. non-rigid folding; why only rigid folding preserves the isometry we rely on.
- Maekawa's theorem: $|M - V| = 2$ at every interior vertex of a flat-foldable pattern.
- Kawasaki's theorem: alternating angle sums at an interior vertex are equal.
- Boxed aside, one page, labelled background: the Huzita–Hatori axioms.
- Piecewise-flat surfaces sit *outside* the smoothness assumptions of manifold-learning theory. Stated as realism, not as a caveat — real data manifolds have creases too.